Problem-Solving, Ratios, Rates & Data

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $\frac{3}{x} = \frac{5}{20}$, what is x?

Answer: 12. Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $5x = 60 \Rightarrow x = 12$. Therefore, the verified correct option is 12.

2 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A quantity $f_0 = 1800$ increases by 20% to f_1 and then decreases by 25% to f_2. What is the final quantity $x = f_2$?

Answer: 60. Computing $1800 \times (1 + 0.20) \times (1 - 0.25) = 1620$.

3 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $\frac{9}{2} = \frac{36}{x}$, what is the value of x?

Answer: 8. Cross-multiply: $9x = 2 \cdot 36 \Rightarrow x = 8$

4 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If 6 identical notebooks cost USD 66 in total, what is the price per notebook?

Answer: 11. USD $66 \div 6 = 11$ per notebook.

5 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If 3 identical notebooks cost USD 18 in total, what is the price per notebook?

Answer: 6. Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, USD $18 \div 3 = 6$ per notebook. Therefore, the verified correct option is 6.

6 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A quantity of 390 is decreased by 40%. What is the new value?

Answer: 234. Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $390 - 40\% \times 390 = 234$ Therefore, the verified correct option is 234.

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7 [Foundational] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A quantity $p_0 = 3800$ increases by 10% to p_1 and then decreases by 20% to p_2. What is the final quantity $t = p_2$?

Answer: 260. Computing $3800 \times (1 + 0.10) \times (1 - 0.20) = 3344$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $\frac{a}{b} = \frac{3}{5}$ and $b = 20$, what is a?

Answer: 12. Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $a = \frac{3}{5} \times 20 = 12$ Therefore, the verified correct option is 12.

9 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $\frac{8}{10} = \frac{48}{x}$, what is the value of x?

Answer: 60. Cross-multiply: $8x = 10 \cdot 48 \Rightarrow x = 60$

10 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A quantity of 70 is increased by 50%. What is the new value?

Answer: 105. Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $70 + 50\% \times 70 = 105$ Therefore, the verified correct option is 105.

11 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A car uses 8 liters of fuel to travel 100 km. How many liters to travel 350 km?

Answer: 28. $\frac{8}{100} \times 350 = 28$ liters.

12 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A mixture is made of three ingredients in the ratio $3 : 1 : 1$. If the total mixture weighs 50 grams, how many grams of the first ingredient are used?

Answer: 30. Each part $= \frac{50}{5} = 10$ g. The first ingredient has 30 g.

13 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A mixture is made of three ingredients in the ratio $5 : 6 : 4$. If the total mixture weighs 60 grams, how many grams of the first ingredient are used?

Answer: 20. Each part $= \frac{60}{15} = 4$ g. The first ingredient has 20 g.

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14 [Intermediate] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A mixture is made of three ingredients in the ratio $3:1:3$. If the total mixture weighs 49 grams, how many grams of the third ingredient are used?

Answer: 21. Each part $= \frac{49}{7} = 7$ g. The third ingredient has 21 g.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A car travels 240 miles in 4 hours. How far in 7 hours?

Answer: 420. Rate $= 60$ mph. $60 \times 7 = 420$ miles.

16 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

What is 25% of 420?

Answer: 105. $25\% \times 420 = 0.25 \times 420 = 105$

17 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A quantity of 40 is decreased by 15%. What is the new value?

Answer: 34. Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $40 - 15\% \times 40 = 34$ Therefore, the verified correct option is 34.

18 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A car travels at a constant speed of 30 miles per hour for 7 hours. How far does it travel?

Answer: 210. Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, Distance $= 30 \times 7 = 210$ miles. Therefore, the verified correct option is 210.

19 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A line of best fit is $y = 8x + 40$ (hours studied vs. score). How many hours to reach a score of 88?

Answer: 6. Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $88 = 8x + 40 \Rightarrow x = 6$ hours. Therefore, the verified correct option is 6.

20 [Challenging] • [SPR] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

After a 40% decrease, a price is USD 30. What was the original price?

Answer: 50. Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $x(1 - 0.4) = 30 \Rightarrow x = 50$ Therefore, the verified correct option is 50.

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Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A price decreases from USD 80 to USD 60. What is the percentage decrease?

a. 20%

b. 25% ✓

c. 33%

d. 40%

Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $\frac{20}{80} \times 100 = 25\%$ Therefore, the verified correct option is 25%.

22 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A recipe that serves 6 people calls for 10 cups of flour. How many cups of flour are needed to serve 19 people?

a. 95/3 ✓

b. 1903/60

c. 383/12

d. 377/12

$\frac{10}{6} = \frac{x}{19} \Rightarrow x = \frac{10 \times 19}{6} = 95/3$ cups.

23 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A store sells 4 apples for USD 3.00. At this rate, what is the cost of 12 apples?

a. USD 6.00

b. USD 8.00

c. **USD** 9.00 ✓

d. USD 12.00

12 apples is 3 times 4 apples: $3 \times \mathit{USD}3.00 = \mathit{USD}9.00$.

24 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

What is 35% of 200?

a. 60

b. 70 ✓

c. 80

d. 90

Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $0.35 \times 200 = 70$. Therefore, the verified correct option is 70.

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25 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A jacket originally priced at USD 60 is on sale for 20% off. What is the discount amount in dollars?

a. USD 10

b. **USD 12** ✓

c. USD 15

d. USD 48

Discount = $0.20 \times 60 = USD12$.

26 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A recipe requires 2 cups of sugar for every 3 cups of flour. If 9 cups of flour are used, how many cups of sugar are needed?

a. 4

b. 6 ✓

c. 8

d. 9

$\frac{2}{3} = \frac{s}{9} \Rightarrow s = 6$ cups of sugar.

27 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A map scale indicates that 1 inch represents 25 miles. If two cities are 3.2 inches apart on the map, what is the actual distance between them?

a. 64 miles

b. 75 miles

c. **80 miles** ✓

d. 85 miles

Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $3.2 \times 25 = 80$ miles. Therefore, the verified correct option is 80 miles.

28 [Foundational] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If x is directly proportional to y, and $x = 14$ when $y = 2$, what is x when $y = 5$?

a. 25

b. 30

c. 35 ✓

d. 40

Constant of proportionality $k = x/y = 14/2 = 7$. When $y = 5$, $x = 7 \times 5 = 35$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

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29 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

The scatter plot shows the relationship between study hours and scores. Which best describes the correlation?

a. Strong negative

b. Weak negative

c. No correlation

d. **Strong positive** ✓

As study hours increase, scores consistently increase. The points follow a clear upward trend — strong positive correlation.

30 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A recipe that serves 6 people calls for 8 cups of flour. How many cups of flour are needed to serve 18 people?

a. 24 ✓

b. 14

c. 30

d. 31

$\frac{8}{6} = \frac{x}{18} \Rightarrow x = \frac{8 \times 18}{6} = 24$ cups.

31 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

What is 15% of 80?

a. 8

b. 12 ✓

c. 15

d. 16

Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $0.15 \times 80 = 12$. Therefore, the verified correct option is 12.

32 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A cyclist travels at a constant speed of 12 miles per hour. How many miles does the cyclist travel in 3.5 hours?

a. 36

b. 40

c. 42 ✓

d. 48

$\text{Distance} = \text{speed} \times \text{time} = 12 \times 3.5 = 42$ miles.

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33 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A class of 30 students has 18 girls and 12 boys. What percentage of the class is boys?

a. 30%

b. 36%

c. 40% ✓

d. 45%

$\frac{12}{30} = \frac{2}{5} = 40\%$.

34 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If a car uses 4 gallons of gasoline to travel 120 miles, how many gallons would it use to travel 300 miles at the same rate?

a. 8

b. 10 ✓

c. 12

d. 15

Fuel economy is $120/4 = 30$ mpg. For 300 miles: $300/30 = 10$ gallons.

35 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If a 12-ounce beverage contains 150 calories, how many calories are in an 8-ounce serving of the same beverage?

a. 90

b. 100 ✓

c. 110

d. 120

Calories per ounce = $150/12 = 12.5$. For 8 ounces: $12.5 \times 8 = 100$ calories.

36 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A car travels 180 miles in 3 hours. At this same rate, how far will it travel in 5 hours?

a. 240 miles

b. 270 miles

c. **300 miles** ✓

d. 360 miles

Speed $= 180/3 = 60$ mph. In 5 hours: $60 \times 5 = 300$ miles.

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37 [Intermediate] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

There are 1,000 milliliters in 1 liter. How many milliliters are in 3.4 liters?

a. 340 mL

b. **3,400 mL** ✓

c. 34,000 mL

d. 34 mL

$3.4 \times 1000 = 3,400$ mL.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If 5 meters is approximately 16.4 feet, approximately how many feet are in 15 meters?

a. 32.8

b. 41.0

c. 49.2 ✓

d. 54.6

15 meters $= 3 \times 5$ meters $\approx 3 \times 16.4 = 49.2$ feet.

39 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

An item's price increased from USD 50 to USD 65. What was the percent increase?

a. 15%

b. 25%

c. 30% ✓

d. 35%

Percent increase $= \frac{65-50}{50} \times 100\% = \frac{15}{50} \times 100\% = 30\%$.

40 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A runner completes a 5-kilometer race in 25 minutes. What is the runner's average speed in kilometers per hour?

a. 10 km/h

b. **12 km/h** ✓

c. 15 km/h

d. 20 km/h

25 minutes is $25/60 = 5/12$ hours. Speed $= 5/(5/12) = 12$ km/h.

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41 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

An electronic device is discounted from USD 200 to USD 160. What is the percent discount?

a. 15%

b. 20% ✓

c. 25%

d. 40%

$\frac{200-160}{200} \times 100\% = \frac{40}{200} \times 100\% = 20\%$.

42 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

What number is 40% of 250?

a. 80

b. 90

c. 100 ✓

d. 120

Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $0.40 \times 250 = 100$.

Therefore, the verified correct option is 100.

43 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

If $\frac{a}{15} = \frac{4}{5}$, what is the value of a?

a. 10

b. 12 ✓

c. 14

d. 16

$a = 15 \times \frac{4}{5} = 12$.

44 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

A store sells notebooks in packs of 5 for USD 6. How much would 25 notebooks cost at this rate?

a. USD 24

b. **USD** 30 ✓

c. USD 35

d. USD 40

25 notebooks is 5 packs: $5 \times USD6 = USD30$.

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45 [Challenging] • [MCQ] • Target: Ratios, Unit Conversions, Percentages & Scatterplots

What is 8% of USD 450?

a. USD 32

b. **USD** 36 ✓

c. USD 40

d. USD 45

Applying Ratios, Rates, Proportions, Percentages & Scatterplots principles, $0.08 \times 450 = USD36$.

Therefore, the verified correct option is USD 36.

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